Conversion rate challenge

CDSD certification Bloc 3

Cyril Braguy · Date 07/11/2026

Introduction et objectif

Introduction

This is a machine learning competition.

www.datascienceweekly.org is a famous newsletter on which anyone can register to receive weekly news about data science.

The data scientists creators of the newsletter would like to understand better the behaviour of the visitors regarding conversion to newsletter by building a predicting model from a few information.

They organised a competition by sharing datasets, and decided to evaluate results with f1 score..

Goal of the project

1. EDA ; preprocessing and train the baseline with the file *data_train.csv*

2. improve your model's **f1-score** on your test set: (feature engineering, feature selection, regularization, non-linear models, hyperparameter optimization by grid search, etc...)

3. make some predictions with *data_test.csv (*and dump it to your teacher/TA 🤓). several submissions are made with different models !

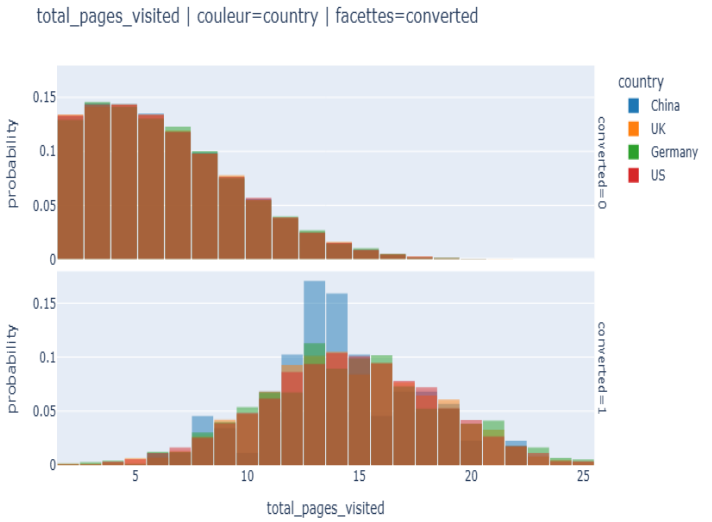
4. Analyze your best model's parameters.
Propose actions that would help to improve the newsletter's conversion rate, Make recommendations to the team

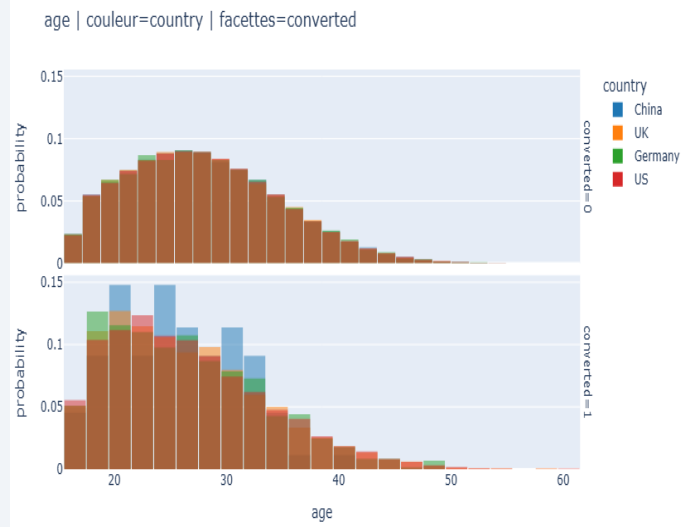
1 - EDA Exploratory Data Analysis

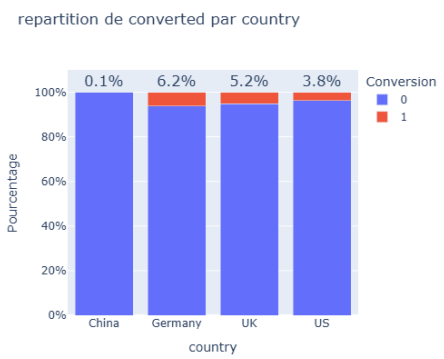
	country	age	new_user	source	total_pages_visited	converted
count	284580	284580.000000	284580.000000	284580	284580.000000	284580.000000
unique	4	NaN	NaN	3	NaN	NaN
top	US	NaN	NaN	Seo	NaN	NaN
freq	160124	NaN	NaN	139477	NaN	NaN
mean	NaN	30.564203	0.685452	NaN	4.873252	0.032258
std	NaN	8.266789	0.464336	NaN	3.341995	0.176685
min	NaN	17.000000	0.000000	NaN	1.000000	0.000000
25%	NaN	24.000000	0.000000	NaN	2.000000	0.000000
50%	NaN	30.000000	1.000000	NaN	4.000000	0.000000
75%	NaN	36.000000	1.000000	NaN	7.000000	0.000000
max	NaN	123.000000	1.000000	NaN	29.000000	1.000000

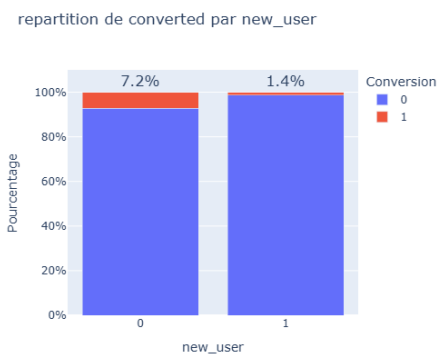
Quite big Dataset: 280 k lines

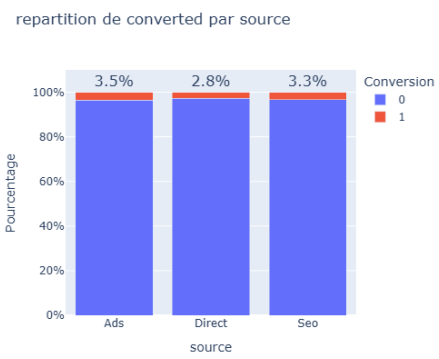
- 5 features & 1 target label (converted)
- Average conversion rate : 3.22%
- Clean data from outliers : keep age <= 75 total_pages_visited <= 25
- Age and pages_visited look to have an impact on conversion
- China repartition curves looks different and strange compared to other countries
- **=> main impacting feature is total_pages_visited**
- Then **age** and **new_user**
- Slight impact from **country (especially China)** and **source**











1 - train baseline and first level of optimisation

First model baseline is using a **logistic regression with one feature : total_pages_visited**
then with all features

Then gridsearch is tested with several features and others models

Best result at this simple stage is given by

Random Forest with with gridsearch with all features gives a pretty good result

Nr	Model	Remove outliers	Features	F1 train	F1 test
0	Logistic	0	**total_pages_visited**	**0.69**	
1	Logistic		all	**0.76**	**0.757**
2	Logistic		All - age	0.74	
3	Logistic + gridsearch		All	0.7649	0.7568
4	Random Forest+gs		All	0.778	0.7563
5	**Random Forest+gs**		All	**0.7727**	**0.7609**
8	Bagging +logreg			0.7411	0.7426
11	Adaboost + logistic			0.6326	0.6372
12.2a	XGBoost		All	**0.7666**	**0.7608**

2- Improve model : Feature engineering & feature selection

improve the model's f1-score on our test set:

- A specific pipeline is built to allow automatic searching : FeatureEngineer → Preprocessor → Classifier

Several tests are launched to evaluate effect of:

- feature engineering : no significant effect
- regularization : small effect
- non-linear models: random forest, XGBoost,
- hyperparameter optimization by grid search : sensible effect

XGBoost is identified as the best performer model

- Feature importance is also used to best locate most efficient features and re-run the model fit

Example on model 16a XGBoost:

num__total_pages_visited	**0.650471**
cat__country_China	**0.163329**
num__new_user	**0.13485**
num__age	**0.025033**
cat__country_US	**0.016332**
cat__country_UK	**0.006276**
cat__country_Germany	**0.003708**

Nr	Model	params	Features	F1 train	F1 test
12.2a	XGBoost		All	**0.7666**	**0.7608**
12.2b	XGBoost	Threshold on Xfull	All	0.7722	
13	XGBoost		pages, country & new_user	0.7504	0.7404
14a	XGBoost		country, new_user, total_pages_visited, age	0.767167	0.759203
14b	XGBoost	Threshold on Xfull		0.77047	
15		+ stacking & bootstrap		0.7718	0.7608
16	XGBoost	rmoutl	country, new_user, total_pages_visited, age + pages*new_user	0.7671	0. 75920

3 - make some predictions with *data_test.csv and submit*

Several models and parameters are tested

Predictions are computed with *data_test.csv (*and dump to our teacher/TA 🤓 Pietro Scalisi).

Some small variations are also tested :

- Train model on X_train -> *f1-score*
- Test model on X_test -> *f1-score*
- Re-train model on X_full = [X_train, X_test]
- Possible to recompute best threshold on X_full

Models 14a & 16a give equivalent results with my best f1-score at a pretty good level

But just 1% margin better than model N°1 logistic regression

CLASSEMENT:

	Rank	Studente	F1	Precision	Recall
0	1	conversion_data_test_predictions_Cyril_Braguyscen14a	0.763240	0.811258	0.720588
1	2	conversion_data_test_predictions_Cyril_Braguyscen16a	0.763240	0.811258	0.720588
2	3	conversion_data_test_predictions_Cyril_Braguyscen16bisa	0.763240	0.811258	0.720588
3	4	conversion_data_test_predictions_Cyril_Braguyscen12.2b	0.762800	0.816555	0.715686
4	5	conversion_data_test_predictions_Cyril_Braguyscen14b	0.761506	0.816143	0.713725
5	6	conversion_data_test_predictions_Cyril_Braguyscen16b	0.761506	0.816143	0.713725
6	7	conversion_data_test_predictions_Cyril_Braguyscen16bisb	0.761506	0.816143	0.713725
7	8	submission_final_Fatimatou	0.761411	0.808370	0.719608
8	9	conversion_data_test_predictions_Cyril_Braguyscen12.2a	0.759853	0.818800	0.708824
9	10	conversion_data_test_predictions_Sandra_RACHDI_V6	0.759073	0.781120	0.738235

4. Analyze best model's parameters.

- Best model is version 14a / 16a :
- Model : **XGBoost**

Features : country, new_user, total_pages_visited, age + rmoutl
Feature engineering : pages*new_user

Hyperparameters : 'clf__colsample_bytree': 0.85, learning_rate': 0.05,

max_depth': 4,

n_estimators': 200,

subsample': 0.85

- F1-train : 0.7671 ; f1-test : 0.75920
- best_threshold: 0.41 ; f1_test_opt : 0.76833
- F1 data_test : 0.763240 ;

Feature importance:

num__total_pages_visited	0.650471
cat__country_China	0.163329
num__new_user	0.13485
num__age	0.025033
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Conclusion :
Conversion rate is highly linked on number of pages visited, then on second order if user is from China or not, and if user is new.
Then on a third order : age (young people convert more)

Actions Proposal to help to improve the newsletter's conversion rate :

- Implement a different navigation experience on website in function of **conversion ML prediction** to propose some conversion messages differently from ML prediction.
- improve the website / information for low converting people : those from China ; not new users, older people,